

YOTRACKEZ — Technical Documentation

AI-Powered Nutrition & Calorie Tracking Web Application (Vercel Live Deployment)

Production Live Build

1. System Overview

YOTRACKEZ is an intelligent nutrition and wellness web application designed to track meals, compute real-time macronutrient balances, and provide dynamic companion avatar feedback. The user captures or uploads a photo of food, which is securely forwarded through a Vercel serverless proxy to Google's Gemini Vision model for instant, structured nutritional breakdown.

2. Tech Stack Summary (Live Deployed Version)

Layer	Technology	Role & Details
Frontend Web Client	Flutter 3.x (Dart Web)	High-performance reactive Web SPA deployed directly to Vercel edge hosting.
AI / Vision Model	Google Gemini 2.0 Flash (<code>gemini-2.0-flash</code>)	Multimodal vision reasoning for food identification, ingredient decomposition, and macronutrient estimation.
Backend API Proxy	Vercel Serverless Function (Node.js)	Secure Edge endpoint (<code>/api/analyze</code>) that proxies image payload to Gemini and secures private API credentials.
Client Persistence	SharedPreferences & IndexedDB / Web Storage	Stores user biometric profiles, daily calorie targets, and logged meal history locally.
Interactive Avatar	Rive 2D/2.5D Vector Runtime	Dynamic companion character adapting across 5 visual wellness states according to net calorie balance.

Layer	Technology	Role & Details
Typography & UI	Google Fonts (Outfit)	Modern dark-mode aesthetic with custom design tokens (<code>#07080F</code> / <code>#7B6FFF</code>).

Note on Production Scope: Hardware-dependent mobile modules (hardware pedometer accelerometer and native camera barcode scanning) are intentionally omitted from this web build to ensure zero-dependency cross-browser compatibility on Vercel.

3. Architecture Flow

- 1. **Image Ingestion:** The user snaps or uploads a meal photo in the web app.
- 2. **Preprocessing & Serialization:** Client compresses image into Base64 format and issues a `POST` request to `/api/analyze` .
- 3. **Secure Proxy Execution:** Vercel serverless handler forwards the request to Google Gemini Vision API with structured JSON schema constraints.
- 4. **Nutritional Parse & Render:** Gemini outputs calories, carbs, protein, fats, fiber, and ingredient lists. The client parses and presents interactive macro rings.
- 5. **Companion Avatar Adjustment:** Daily net calorie balance recalculates, automatically updating the companion avatar's mood and body state.

4. Core Application Modules & Services

Service / Module	File Path	Key Responsibility
GeminiFoodService	lib/services/gemini_food_service.dart	Transmits image payload to the Vercel backend proxy and deserializes the structured nutrition response.
ImagePreprocessor	lib/services/image_preprocessor.dart	Downsamples and optimizes

Service / Module	File Path	Key Responsibility
		image resolution for swift web upload without quality loss.
MealDbService	lib/services/meal_db_service.dart	Manages meal logging, timestamps, calorie tallying, and persistent meal history.
CalorieBalanceEngine	lib/services/calorie_balance_engine.dart	Computes net calorie balance (calories consumed vs. expenditure goals) to adjust daily health standing.
BmrCalculator	lib/services/bmr_calculator.dart	Calculates Basal Metabolic Rate via standard Mifflin-St Jeor formula based on age, height, weight, and gender.
RecommendationService	lib/services/recommendation_service.dart	Generates contextual AI wellness advice and

Service / Module	File Path	Key Responsibility
		micro-tips based on daily macro ratios.

5. Deployed Application Screens

Screen	Primary Functionality
Splash Screen	Initializes brand assets, verifies cached user profile, and routes to home.
Onboarding Screen	Captures user biometrics (height, weight, age, fitness goal) to compute initial calorie allowances.
Dashboard / Home Screen	Central hub displaying daily calorie ring, macro distribution bars, and the animated companion character.
Result Screen	Detailed nutritional breakdown showing serving sizes, micro/macro nutrients, and ingredient estimates.
Meal History Screen	Chronological ledger of logged meals grouped by calendar date and meal type (Breakfast, Lunch, Dinner, Snack).
Profile & Settings	Interface for customizing personal health goals, updating weight/height, and calibrating calorie goals.

6. Backend API Specification

```
POST /api/analyze
Content-Type: application/json

Request Payload:
{
  "image": "<base64_encoded_jpeg_or_png>",
  "mimeType": "image/jpeg"
}

Successful Response:
{
```

```
"food_name": "Grilled Chicken Salad",
"serving_size": "1 bowl (approx 350g)",
"calories": 420.0,
"carbs": { "name": "Carbohydrates", "amount": 14.0, "unit": "g", "daily_percent": 5.0 },
"protein": { "name": "Protein", "amount": 48.0, "unit": "g", "daily_percent": 96.0 },
"fat": { "name": "Fat", "amount": 18.0, "unit": "g", "daily_percent": 23.0 },
"fiber": { "name": "Fiber", "amount": 6.0, "unit": "g", "daily_percent": 21.0 },
"ingredients": [
  { "name": "Chicken Breast", "amount": "200g", "calories": 260, "protein": 40, "fat": 10 },
  { "name": "Mixed Greens & Olive Oil", "amount": "150g", "calories": 160, "protein": 5, "fat": 10 },
],
"health_tip": "High in lean protein, optimal for muscle recovery and satiety."
}
```

